

Q RTV 2014

In a class containing 50 students, 15 play Tennis, 20 play Cricket and 20 play Hockey, 3 play Tennis and Cricket, 6 play Cricket and Hockey and 5 play Tennis and Hockey. 7 play no game at all. How many play Cricket, Tennis and Hockey.

Solⁿ $\rightarrow$ Let the sets of Tennis, Cricket and Hockey players be denoted by T, C and H respectively.

$$n(U) = 50$$

$$n(T) = 15, \quad n(C) = 20, \quad n(H) = 20$$

$$n(T \cap C) = 3, \quad n(C \cap H) = 6, \quad n(T \cap H) = 5$$

$$n(T \cup C \cup H)' = 7$$

We want to find $n(T \cap C \cap H)$

Now By addition theorem for three events

$$n(T \cup C \cup H) = n(T) + n(C) + n(H) - n(T \cap C) - n(C \cap H) - n(T \cap H) + n(T \cap C \cap H) \rightarrow \textcircled{1}$$

$$\Rightarrow n(T \cup C \cup H) = 15 + 20 + 20 - 3 - 6 - 5 + n(T \cap C \cap H)$$

$$\Rightarrow n(T \cup C \cup H) = 41 + n(T \cap C \cap H) \longrightarrow (2)$$

But we know that

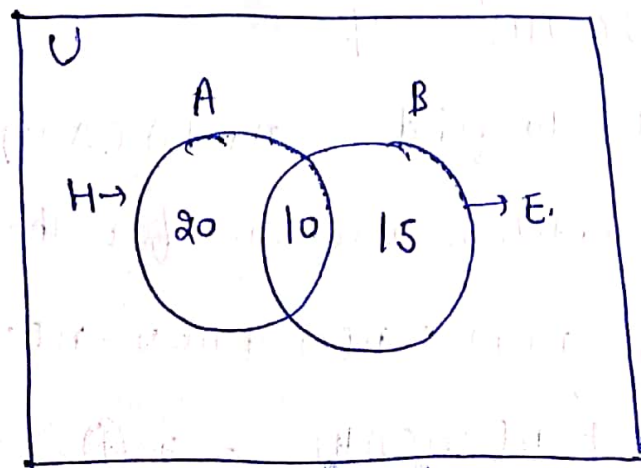
$$n(T \cup C \cup H)' = n(U) - n(T \cup C \cup H)$$

$$\Rightarrow 7 = 50 - [41 + n(T \cap C \cap H)] \quad (\text{From } (2))$$

$$\begin{aligned} \Rightarrow n(T \cap C \cap H) &= 50 - 41 - 7 \\ &= 2 \end{aligned} \quad \text{Ans.}$$

Ex $\rightarrow$ In a class of 50 students, 30 are studying Hindi and 25 English Language and 10 are studying both languages. How many students are studying either language.

Soln



Soln → Let U be the class of 50 students, A ⁽¹⁷⁾ is a set of those students studying Hindi and B be the set of those studying English. We have to find $n(A \cup B)$.

Given that $n(A) = 30$, $n(B) = 25$, $n(A \cap B) = 10$

We have

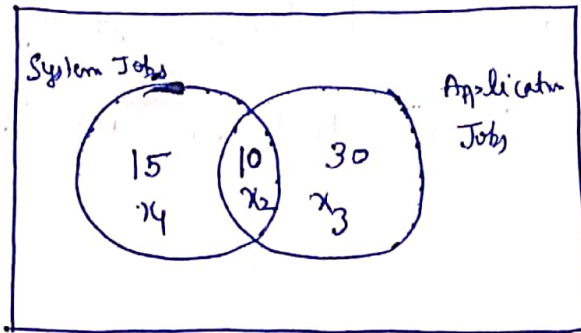
$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$\Rightarrow n(A \cup B) = 30 + 25 - 10 \\ = 45 \text{ } \underline{A}$$

Therefore the number of students whose studying either English or Hindi is 45 A.

Ex → A computer company must hire 25 programmers to handle systems programming jobs and 40 for the application programming. of the hired persons, 10 will have to do jobs of both types. find how many programmers must be hired.

Soln →



According to the diagram

$$x_1 + x_2 = 25 \rightarrow \textcircled{1}$$

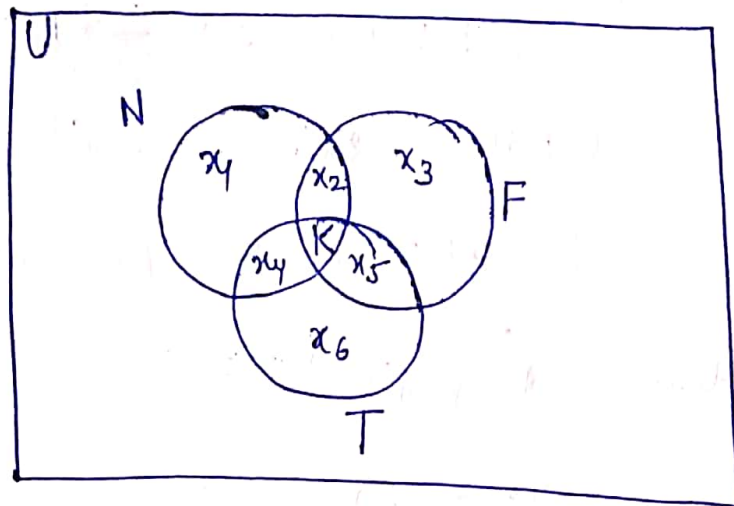
$$x_2 + x_3 = 40 \rightarrow \textcircled{2}$$

$$x_2 = 10 \rightarrow \textcircled{3}$$

$$x_1 = 15, \quad x_3 = 30$$

$$\therefore \text{Total} = x_1 + x_2 + x_3 = 55 \quad \underline{\underline{A}}$$

Ex → In a survey of 60 people it was found that 25 read News week, 26 read Time and 26 read Magazine fortune. Also 9 read both News week and Fortune, 11 read News week and Time and 8 read both Time and Fortune. If 8 read none of the three magazines, determine the number of peoples who read exactly one magazine.



According to the venn diagram.

$$(x_1 + x_3 + x_6) + (x_2 + x_4 + x_5) + K = 52 \rightarrow (1)$$

$$x_1 + (x_2 + x_4) + K = 25 \rightarrow (2)$$

$$x_2 + (x_3 + x_5) + K = 26 \rightarrow (3)$$

$$x_4 + (x_5 + x_6) + K = 26 \rightarrow (4)$$

$$x_2 + K = 9 \rightarrow (5)$$

$$x_4 + K = 11 \rightarrow (6)$$

$$x_5 + K = 8 \rightarrow (7)$$

Adding (2), (3) & (4) we get

$$(x_1 + x_3 + x_6) + 2(x_2 + x_4 + x_5) = 77 \rightarrow (8)$$

and Adding (5), (6) & (7)

$$(x_2 + x_4 + x_5) + 3K = 28 \rightarrow (9)$$

Let $x_1 + x_3 + x_6 = p$ and $x_2 + x_4 + x_5 = q$

then From (1)

$$p + q + k = 52 \rightarrow (10)$$

$$q + 3k = 28 \rightarrow (11)$$

$$\text{and } p + 2q + 3k = 77 \rightarrow (12)$$

Now adding (10) & (11)

$$p + 2q + 4k = 80 \rightarrow (13)$$

and subtracting (13) by (12)

$$\boxed{k=3} \rightarrow (14)$$

put the value of k in (11)

$$\text{we get } q = 28 - 9 = 19$$

$$\text{and } p = 52 - 3 - 19 = 52 - 22 = 30$$

$$\text{hence } x_1 + x_3 + x_6 = p = 30$$

Since those who read news week only = x_1

Those who read Time only = x_3

Those who read Fortune only = x_6

Those who read only one magazine

$$= x_1 + x_3 + x_6 = p = 30$$

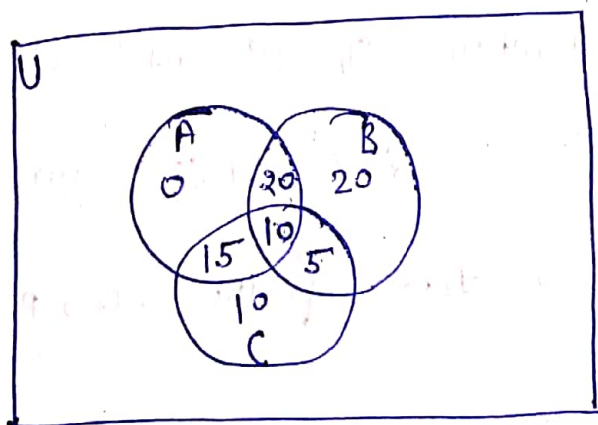
Ex → In a town 45% read magazine A, 55% read magazine B, 40% read magazine C, 30% read magazines A and B, 15% read magazines B and C, 25% read C and A, 10% read all the three magazines. Find what percentage do not read any magazine? What percentage reads exactly two of the magazines?

Solⁿ → Let A, B and C denotes the set of all those who read magazines A, B and C respectively.

Then ~~$n(A) = 45$~~ , $n(A) = 45$, $n(B) = 55$, $n(C) = 40$

$n(A \cap B) = 30$, $n(B \cap C) = 15$, $n(A \cap C) = 25$,

$n(A \cap B \cap C) = 10$, $n(U) = 100$



From the diagram it is easy to understand that, The number of persons who read only A and B but not C = 20

(15)

The number of persons who read B and C but not A
 $= 5$

The number of persons who read C and A but not B
 $= 15$

The number of persons who read only A $= 0$

The number of persons who read only B $= 20$

The number of persons who read only C $= 10$

The number of persons who read at least one
magazine $= 0 + 20 + 10 + 20 + 15 + 5 + 10 = 80$

So, those who do not read any magazine are
20 in number.

So 20% do not read any newspaper (magazine)

and The number of persons who read exactly
two magazines $= 20 + 5 + 15 = 40$.

40% read two of the newspaper.



(14)

Ex $\Rightarrow$ In a group of 52 persons 16 drink tea but not coffee and 33 drink tea. Find

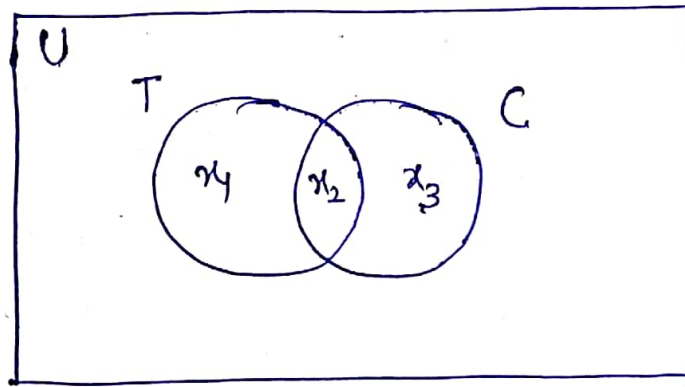
(i) How many drink tea and coffee both

(ii) How many drink coffee but not tea.

Solⁿ $\Rightarrow$ Let U be the set of 52 persons. Let

T denotes the sets of persons who drink tea and

C denotes the set of persons who drink coffee.



$n(T-C) = x_1 =$ number of persons who drink Tea but not coffee.

$n(T \cap C) =$ no. of persons who drink Coffee and Tea both
 $= x_2$

$n(C-T) =$ no of persons who drink Coffee but not tea
 $= x_3$

$$\Rightarrow n(T-C) = x_1, \quad n(T \cap C) = x_2, \quad n(C-T) = x_3$$

$$\Rightarrow x_1 + x_2 + x_3 = 52 \rightarrow \textcircled{1}$$

$$\underline{x_1 = 16} \rightarrow (2)$$

$$\Rightarrow x_1 + x_2 = 33 \rightarrow (3)$$

$$\therefore x_2 = 33 - 16 = 17 \quad \text{and} \quad x_3 = 52 - 33 = 19$$

(i) Those who drink tea and coffee both = $x_2 = 17$

(ii) Those who drink coffee but not tea = $x_3 = 19$

A

Set theory :

(1) Identity laws

(a) $A \cup \phi = A$

(b) $A \cap U = A$

(2) Bound laws

(a) $A \cup U = U$

(b) $A \cap \phi = \phi$

(3) Absorption laws

(a) $A \cup (A \cap B) = A$

(b) $A \cap (A \cup B) = A$

(4) Involution laws

(a) $A \cup A' = U$

(b) $(A')' = A$

Duality: Two formulae's / Identities E & E^* in set algebra is said to be dual of each other if either one can be obtained from the other by replacing $\cup$ by $\cap$, $\cap$ by ' $\cup$ ', U by ϕ and ϕ by U .

For ex 1(a), 1(b) are dual to each other and so on.

(A)

Q)) Write the dual of the each of following set equation.

i) $(A \cap U) \cap (\phi \cup A') = \phi$

ii) $(A \cup B \cup C)' = (A \cup C)' \cap (A \cup B)'$

Soln

i) $(A \cup \phi) \cup (U \cap A') = U$

ii) $(A \cap B \cap C)' = (A \cap C)' \cup (A \cap B)'$

Questions based on Three sets A, B & C
(operations)

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) \\ - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$$

Q1) In a group of 200 people, each of whom is atleast accountant or management consultant or sales manager, it was found that 80 are accountants, 110 are management consultant and 130 are sales managers, 25 are accountants as well as sales managers, 70 are management consultants as well as sales managers, 10 are accountants, managements as well as sales manggery. Find the number of those people who are accountants as well as management consultants but not sales managers.

Soln: Let A, M and S denote the set of accountants, mangement consultants & sales managers.

(C)

∴ It is given that

$$n(A) = 80, \quad n(M) = 110, \quad n(S) = 130$$

$$n(ANS) = 25, \quad n(MNS) = 70 \text{ and } n(ANMS) = 10$$

By inclusion - Exclusion principle

$$n(A \cup M \cup S) = n(A) + n(M) + n(S) - n(AM) - n(MS) - n(AS) + n(ANMS)$$

$$\Rightarrow 200 = 80 + 110 + 130 - n(AM) - 70 - 25 + 10$$

$$\Rightarrow n(AM) = 35$$

Now

$$n(ANMS') = n(AM) - n(ANMS)$$

$$= 35 - 10 = 25 \quad \underline{\underline{Ar}}$$

9)) A TV survey show that 60 percent people see programme A, 50 percent see programme B, 50 percent see programme C, 30 percent see programme A and B, 20 percent see programme B and C, 30 percent see programme A and C

and 10 percent do not see any programme

Find (i) What percent see programme A, B and C?

(ii) What percent see exactly two programmes?

(iii) What percent see programme A only?

Soln: Suppose X , Y and Z denotes the set of people see programme A, B and C.

$$n(X) = 60, \quad n(Y) = 50, \quad n(Z) = 50,$$

$$n(X \cap Y) = 30, \quad n(Y \cap Z) = 20, \quad n(X \cap Z) = 30$$

$$n(X^c \cap Y^c \cap Z^c) = 10 = n(X \cup Y \cup Z)^c$$

$$(a) \quad n(X \cup Y \cup Z) + n(X \cup Y \cup Z)^c = 100$$

Here we find $n(X \cap Y \cap Z) = ?$

$$\text{now } n(X \cup Y \cup Z) = 100 - 10 = 90 \longrightarrow (1)$$

also

$$\begin{aligned} n(X \cup Y \cup Z) &= n(X) + n(Y) + n(Z) - n(X \cap Y) \\ &\quad - n(Y \cap Z) - n(X \cap Z) + n(X \cap Y \cap Z) \end{aligned}$$

$$\Rightarrow 90 = 60 + 50 + 50 - 30 - 20 - 30 + n(X \cap Y \cap Z)$$

$$\Rightarrow n(X \cap Y \cap Z) = 10$$

10 percent see programme A, B and C.

(D)

(b) Here we have to find for exactly two programmes? as

$$n(X \cap Y \cap Z^c), n(Y \cap Z \cap X^c), n(X \cap Z \cap Y^c)$$

now

$$n(X \cap Y \cap Z^c) + n(X \cap Y \cap Z) = n(X \cap Y)$$

$$\Rightarrow n(X \cap Y \cap Z^c) = n(X \cap Y) - n(X \cap Y \cap Z) \\ = 30 - 10 = 20$$

also

$$n(Y \cap Z \cap X^c) = n(Y \cap Z) - n(X \cap Y \cap Z) \\ = 20 - 10 = 10$$

and

$$n(X \cap Z \cap Y^c) = n(X \cap Z) - n(X \cap Y \cap Z) \\ = 30 - 10 = 20$$

$\therefore$ the total percentage of people to see programme ~~A, B and C~~ exactly two programmes

$$= 20 + 20 + 10 = 50.$$

$$(c) \quad n(X \cap Y^c \cap Z^c) = n(X) - n(X \cap Y) - n(X \cap Z) \\ + n(X \cap Y \cap Z)$$

$$= -30 - 30 + 60 + 10 = 10 \quad \underline{\underline{Ans}}$$

Q) of a group of 20 persons, 10 are interested in music, 7 are interested in photography, and 4 like swimming; furthermore 4 are interested in both music and photography, 3 are interested in both music and swimming, 2 are interested in both photography and swimming and one is interested in music, photography and swimming. How many interested in photography but not in music and swimming.

Soln: Suppose M , P and S denote the set of persons interested in music, ~~photograph~~ photography and swimming respectively.

$$n(M) = 10, \quad n(P) = 7, \quad n(S) = 4$$

$$n(M \cap P) = 4, \quad n(M \cap S) = 3, \quad n(P \cap S) = 2$$

$$\& \quad n(M \cap P \cap S) = 1$$

(E)

$$n(P \cap M^c \cap S^c) = n[P \cap (M \cup S)^c]$$

$$= n(P) - n(P \cap M) - n(P \cap S) + n(P \cap M \cap S)$$

$$= 7 - 4 - 2 + 1 = 2 \quad \underline{\text{Ans}}$$

Q1) Determine the number of integers between 1 and 250 that are divisible by any of the integers 2, 3, 5 and 7.

Soln Let A_1, A_2, A_3 & A_4 denote the set of integers between 1 and 250 that are divisible by 2, 3, 5 & 7 respectively.

$$n(A_1) = \frac{250}{2} = 125, \quad n(A_2) = \frac{250}{3} = 83,$$

$$n(A_3) = \frac{250}{4} = 62, \quad n(A_4) = \frac{250}{7} = 35$$

$$\text{Thus } n(A_1 \cap A_2) = 250 / 2 \times 3 = 41$$

$$n(A_1 \cap A_3) = 250 / 2 \times 5 = 25$$

$$n(A_1 \cap A_4) = 250 / 2 \times 7 = 17$$

$$n(A_2 \cap A_3) = 250 / 3 \times 5 = 16$$

$$n(A_2 \cap A_4) = 250 / 3 \times 7 = 11$$

$$n(A_3 \cap A_4) = 250 / 5 \times 7 = 7$$

$$n(A_1 \cap A_2 \cap A_3) = 250 / 2 \times 3 \times 5 = 8$$

$$n(A_1 \cap A_2 \cap A_4) = 250 / 2 \times 3 \times 7 = 5$$

$$n(A_1 \cap A_3 \cap A_4) = 250 / 2 \times 5 \times 7 = 3$$

$$n(A_2 \cap A_3 \cap A_4) = 250 / 3 \times 5 \times 7 = 2$$

$$\& n(A_1 \cap A_2 \cap A_3 \cap A_4) = 250 / 2 \times 3 \times 5 \times 7 = 1$$

$$\therefore n(A_1 \cup A_2 \cup A_3 \cup A_4)$$

$$= n(A_1) + n(A_2) + n(A_3) + n(A_4) - n(A_1 \cap A_2) -$$

$$- - - - - - - - n(A_1 \cap A_2 \cap A_3 \cap A_4)$$